

RANDOM IMPULSE NOISE REMOVAL USING SPARSE AND LOW RANK DECOMPOSITION OF ANNIHILATING FILTER-BASED HANKEL MATRIX

Kyong Hwan Jin and Jong Chul Ye

Dept. of Bio and Brain Engineering
Korea Advanced Institute of Science and Technology, Republic of Korea

ABSTRACT

Annihilating filter-based low rank Hankel matrix (ALOHA) approach was recently proposed as an intrinsic image model for image inpainting estimation. Based on the observation that smoothness or textures within an image patch are represented as sparse spectral components in the frequency domain, ALOHA exploits the existence of annihilating filters and the associated rank-deficient Hankel matrices in the image domain to estimate the missing pixels. As an extension, here we propose a novel impulse noise removal algorithm using sparse + low rank decomposition of an annihilating filter-based Hankel matrix. This novel approach, what we call robust ALOHA, is inspired by the observation that an image corrupted with impulse noises has intact pixels; so the impulse noises can be modeled as sparse outliers, whereas the underlying image can be still modeled using a low-rank Hankel structured matrix. Numerical results confirm that robust ALOHA has significant performance improvements compared to the state-of-the-art impulse removal algorithms.

Index Terms— ALOHA, annihilating filter, robust principal component analysis, impulse noise denoising

1. INTRODUCTION

Impulse noises occur by malfunctioning of detector pixels in camera or from the missing memory elements in imaging hardware [1]. For the random valued impulse noises (RVIN) which have random values within the dynamic range of an image pixel, the adaptive center-weighted median filter (ACWMF)[2] has been widely used to find the locations of pixels with random valued impulse noises. However, when the density of noise increases, the denoising performance becomes severely degraded. To overcome this weakness, two-phase denoising algorithms with “decision-based filter” or “switching filter” were proposed [3]. These algorithms have two parts: detecting noise pixels by ACWMF or other outlier finding algorithms; and then replacing the detected noise pixels with the estimated values using the total variation[3] or

edge preserving regularizations [4], while keeping noiseless pixels unchanged.

Recently, denoising algorithms for impulse noises using proximal optimizations with non-smooth penalties were proposed [5]. In particular, in the TVL1 (total variation l_1) approach [5], the data fidelity term was measured with the l_1 norm to deal with impulse outliers, whereas the total variation regularization was used as image model. In addition, the related article demonstrated that the wavelet sparsity could be used for finding impulse noises [6]. However, the algorithms often generate distorted information about edge or texture patterns due to the strong TV terms or insufficient information about noise location. On the other hand, to exploit the gain of patch-based approach [7], dictionary based low-rank impulse denoising [8] was also proposed.

Recently, we showed that the edge or texture within an image patch leads to sparsely represented spectrum in the frequency domain, so the sampling theory of signals with finite rate of innovations [9] tells us that there exists an annihilating filter that annihilates the pixel values within the corresponding image patch [10]. Accordingly, the existence of annihilating filter enables us to construct a rank-deficient Hankel structured matrix whose rank is determined by the minimum length annihilating filter [10, 11]. Thanks to this observation, an image patch could be modeled using an annihilating filter-based Hankel structured matrix (ALOHA), and missing image pixels in image inpainting problems can be effectively solved.

One of the most important consequences of ALOHA in the context of impulse noise removal is an observation saying that the construction of Hankel structured matrix is a linear lifting scheme, so the sparse components in local patch of image are also sparse in the lifted Hankel matrix. Therefore, we can exploit a sparse+low rank decomposition of the Hankel structured matrix to decouple the sparse impulse noise components from the underlying image. The new algorithm, what we call robust ALOHA, is applied patch by patch to adapt the local image statistics that have distinct spectral distribution. We are aware that there have been significant progresses on the decomposition of superposed matrix consisting of low-rank and sparse components [12], which is often called Robust Principal Component Analysis (RPCA). However, the

This work was supported by Korea Science and Engineering Foundation under Grant NRF-2014R1A2A1A11052491.

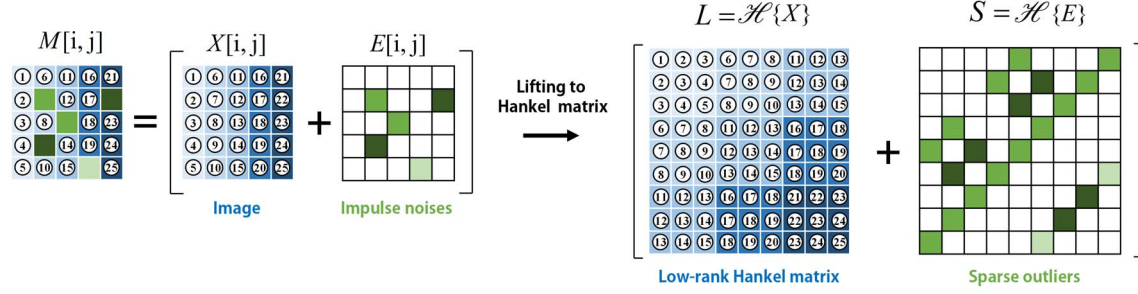


Fig. 1. Sparse + low rank decomposition of Hankel structured matrix from an image patch corrupted with impulse noise. Because a lifting to Hankel structure is linear, the sparse impulse noises are also lifted to sparse outliers.

matrix in RPCA is usually unstructured, whereas the robust ALOHA uses an Hankel structured matrix. This difference makes the robust ALOHA algorithm significantly outperform RPCA approach.

2. SPARSE AND LOW-RANK MODEL FOR IMPULSE NOISES

In our recent work [10], we demonstrate that diffusion and/or Gaussian Markov random field (GMRF) approaches for image modelling are closely related to an annihilating filter relationship from the sampling theory of signals with finite rate of innovations (FRI) [9]. Specifically, we can derive the annihilation relationship between sparse spectrum of local patch and annihilating filter from FRI theory.

Mathematically, when a patch $x(\mathbf{r}) \in \mathbb{R}^2$ has sparse spectral components, we can show that there exists a corresponding annihilating filter in the image domain. For example, if the spectrum of an image patch is described by

$$\hat{x}(\boldsymbol{\omega}) = \sum_{j=0}^{k-1} c_j \delta(\boldsymbol{\omega} - \boldsymbol{\omega}_j), \quad \boldsymbol{\omega} = (\omega_x, \omega_y) \quad (1)$$

where k denotes the number of non-zero spectral components, then it is easy to find an annihilating function $\hat{h}(\boldsymbol{\omega})$ in the spectrum domain that does not overlap with the support of $\hat{x}(\boldsymbol{\omega})$, i.e.

$$\hat{h}(\boldsymbol{\omega})\hat{x}(\boldsymbol{\omega}) = 0 \quad \forall \boldsymbol{\omega}. \quad (2)$$

This implies the existence of the annihilating filter $h(\mathbf{r}) := \mathcal{F}^{-1}\{\hat{h}(\boldsymbol{\omega})\}$ in the image domain:

$$h(\mathbf{r}) * x(\mathbf{r}) = 0. \quad (3)$$

If Fourier measurement data $\hat{x}(\boldsymbol{\omega})$ is discretized at an appropriate Nyquist sampling interval, the corresponding discrete counterpart is given by

$$(h * x)[\mathbf{n}] = \sum_{\mathbf{m}} h[\mathbf{m}]x[\mathbf{n} - \mathbf{m}] = 0, \quad \mathbf{n}, \mathbf{m} \in \mathbb{Z}^2, \quad (4)$$

where the discrete filter $h[\mathbf{n}]$ is now a discrete annihilating filter. Among the various of choices of annihilating function

$\hat{h}(\boldsymbol{\omega})$ that satisfies (2), Vetterli et. al. [9] showed that an annihilating function can be constructed using a finite combination of sinusoids such that the corresponding discrete annihilating filter $h[\mathbf{n}]$ has a finite filter length. In this case, the convolution (4) becomes finite length convolution, and we can exploit (4) in a matrix representation.

Specifically, let $\mathbf{X} \in \mathbb{R}^{N \times M}$ denote the matrix composed of $x[\mathbf{n}]$ such that $\mathbf{X}_{i,j} = x[i, j]$. We also define the discrete filter matrix $\mathbf{H} \in \mathbb{R}^{p \times q}$ such that $\mathbf{H}_{i,j} = h[i, j]$. Then, by removing the boundary data beyond the image patch, we can construct the following matrix equation:

$$\mathcal{H}\{\mathbf{X}\}\overline{\text{VEC}(\mathbf{H})} = \mathbf{0}, \quad (5)$$

where $\text{VEC}(\mathbf{H})$ is the vectorisation of the matrix $\mathbf{H}$ and the overline is the order reversal operation. For the detailed structure of $\mathcal{H}$, please refer [10].

If the underlying signal has k -sparse spectral components, we can further show the following key result [13, 11]:

$$\text{RANK}_{\mathcal{H}}(\mathbf{X}) = k, \quad (6)$$

which implies that as long as the annihilating filter size is bigger than the sparsity level, the rank of the associated Hankel matrix is always equal to the sparsity level. By exploiting this property, our previous work [10] derived an annihilating filter-based low rank Hankel matrix approach for image inpainting. In case of image corrupted by impulse noises, this approach will be recasted into sparse+low rank decomposition for impulse noise removal. When the impulse noises are inserted, the rank of patch based Hankel matrix will be corrupted by this impulse noises which have a role as sparse outliers. However, the sparse outliers are still sparse outliers in lifted Hankel matrix as shown in Fig. 1, so the problem of removal of impulse noises is easily converted into sparse + low-rank decomposition of patch-based Hankel matrix.

3. ROBUST ALOHA FOR SPARSE AND LOW-RANK MODEL

Note that the Hankel structured matrix is determined by the underlying image patch ($\mathbf{X}$) size and the associated annihilating filter ($\mathbf{H}$) size. For given $M \times N$ image patch and $p \times q$

annihilating filter, we now denote the associated spaces for the Hankel matrix as $\mathcal{H}(M, N; p, q)$. Then, for a given noisy image patch $\mathbf{M} \in \mathbb{R}^{M \times N}$ and $p \times q$ annihilating filter size, our impulse noise removal algorithm can be implemented by solving the following sparse + low rank decomposition under the Hankel structure matrix constraint:

$$(P) \quad \begin{aligned} \min_{\mathbf{L}, \mathbf{E}} \quad & \|\mathbf{L}\|_* + \tau \|\mathbf{E}\|_1 \\ \text{subject to} \quad & \mathbf{L} = \mathcal{H}\{\mathbf{X}\} \\ & \mathbf{X} + \mathbf{E} = \mathbf{M}. \end{aligned}$$

where τ denotes balancing parameter between low-rankness and sparsity. More specifically, if a factorized form of nuclear norm relaxation [14] can be applied, then the constraints in (P) can be handled using alternating direction method of multiplier (ADMM) [15, 16].

4. RESULT

4.1. Random Valued Impulse Noise (RVIN) denoising

We performed denoising experiments using randomly distributed random valued impulse noise (RVIN) that corrupts and 40% of the whole image pixels. When $x_{ij} := \mathbf{X}(i, j)$ and $N(x_{ij})$ are the original pixel value at location (i, j) and the contaminated pixel with impulse noise at location (i, j) , respectively, RVIN is described as

$$N(x_{ij}) = \begin{cases} d_{ij} & \text{with probability } p \\ x_{ij} & \text{with probability } 1 - p \end{cases} \quad (7)$$

where d_{ij} is a random number between $[d_{min} \ d_{max}]$ chosen by the uniform random probability density function and p is the proportion of noisy pixels with respect to total pixels. The Barbara image was tested and rescaled to have values between 0 and 1. For comparison, a median filter method (MATLAB built-in function ‘medfilt2’, indicated as MF in the figure) was used as the simplest reference algorithm, and the existing algorithms such as ACWMF[2], and TVL1[17] were also used. For quantitative evaluation, we used the PSNR (peak signal-to-noise ratio). As shown in Fig. 2, denoised image from robust ALOHA showed clear texture pattern on clothes compared with other algorithms as well as high PSNR value.

4.2. Performance comparison with conventional RPCA

To verify the necessity of a lifting to a Hankel matrix, we compared the results from the standard RPCA. Note that the standard RPCA uses an image as they are, without reformulating them into a Hankel structured matrix. For RPCA, we used the software packages provided by the original authors in [18]. As you can see in Fig. 3, denoised image from RPCA for single image denoising was failed in removing impulse noises, and the detailed image structures were distorted. On the other hand, the robust ALOHA provided clearly removed

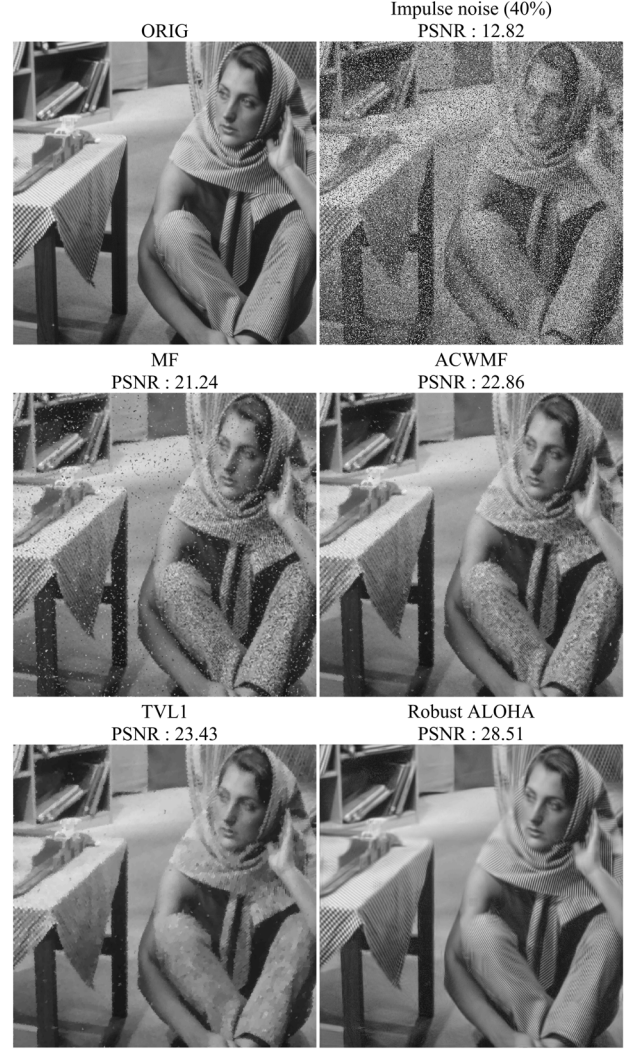


Fig. 2. Reconstructed Barbara images by various methods from 40% random valued impulse noise.

random valued impulse noises. Such a remarkable performance improvement was originated from an image modeling using a low rankness of annihilating filter-based Hankel matrix, which again confirm the robust ALOHA is a superior image model and denoising algorithm for images corrupted with impulse noises.

5. CONCLUSION

In this paper, we proposed a sparse + low rank decomposition of annihilating filter-based Hankel matrices for impulse noise removal. The new algorithm, called robust ALOHA, extends the conventional RPCA approaches by exploiting the spectral domain sparsity of patch and the associated rank deficient Hankel matrix. The robust ALOHA was implemented using ADMM iteration with initialization using LMaFit algorithms.

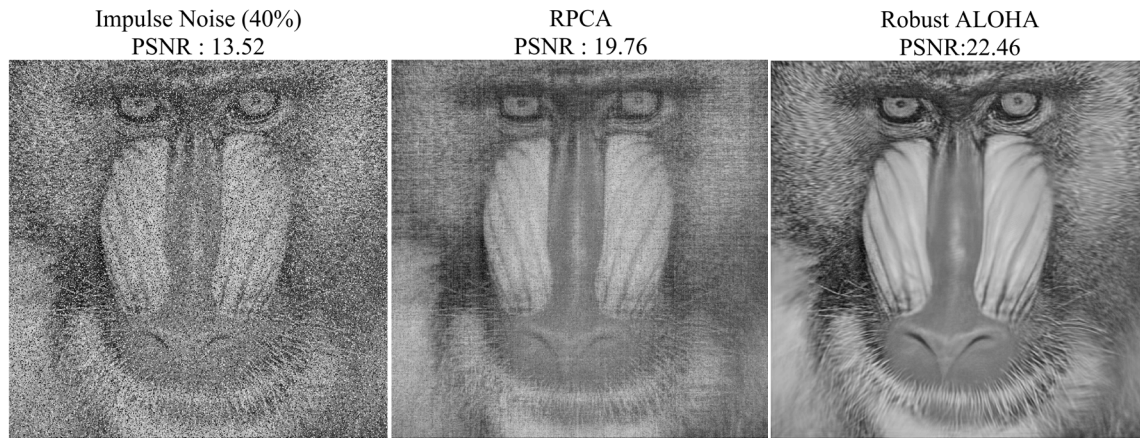


Fig. 3. Comparison with conventional RPCA approach with the proposed method under 40% random valued impulse noises of Baboon image.

The superior performance of the robust ALOHA as well as ALOHA inpainting [10] clearly shows that image modeling using annihilating filter based Hankel matrix is a very powerful tool for many image processing applications.

6. REFERENCES

- [1] Raymond H Chan, Chung-Wa Ho, and Mila Nikolova, "Salt-and-pepper noise removal by median-type noise detectors and detail-preserving regularization," *Image Processing, IEEE Transactions on*, vol. 14, no. 10, pp. 1479–1485, 2005.
- [2] Tao Chen and Hong Ren Wu, "Adaptive impulse detection using center-weighted median filters," *IEEE Signal Process. Lett.*, vol. 8, no. 1, pp. 1–3, 2001.
- [3] Ming Yan, "Restoration of images corrupted by impulse noise and mixed gaussian impulse noise using blind inpainting," *SIAM Journal on Imaging Sciences*, vol. 6, no. 3, pp. 1227–1245, 2013.
- [4] Mila Nikolova, "A variational approach to remove outliers and impulse noise," *Journal of Mathematical Imaging and Vision*, vol. 20, no. 1-2, pp. 99–120, 2004.
- [5] Antonin Chambolle and Thomas Pock, "A first-order primal-dual algorithm for convex problems with applications to imaging," *Journal of Mathematical Imaging and Vision*, vol. 40, no. 1, pp. 120–145, 2011.
- [6] Bin Dong, Hui Ji, Jia Li, Zuowei Shen, and Yuhong Xu, "Wavelet frame based blind image inpainting," *Applied and Computational Harmonic Analysis*, vol. 32, no. 2, pp. 268–279, 2012.
- [7] Kostadin Dabov, Alessandro Foi, Vladimir Katkovnik, and Karen Egiazarian, "Image denoising by sparse 3-D transform-domain collaborative filtering," *IEEE Trans. Image Process.*, vol. 16, no. 8, pp. 2080–2095, 2007.
- [8] Yi Wang, Arthur Szlam, and Gilad Lerman, "Robust locally linear analysis with applications to image denoising and blind inpainting," *SIAM Journal on Imaging Sciences*, vol. 6, no. 1, pp. 526–562, 2013.
- [9] Martin Vetterli, Pina Marziliano, and Thierry Blu, "Sampling signals with finite rate of innovation," *IEEE Trans. Signal Process.*, vol. 50, no. 6, pp. 1417–1428, 2002.
- [10] Kyong Hwan Jin and Jong Chul Ye, "Annihilating filter-based low-rank Hankel matrix approach for image inpainting," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 3498–3511, Nov 2015.
- [11] Yuxin Chen and Yuejie Chi, "Robust spectral compressed sensing via structured matrix completion," *IEEE Trans. Inf. Theory*, vol. 60, no. 10, pp. 6576–6601, 2014.
- [12] Emmanuel J Candès, Xiaodong Li, Yi Ma, and John Wright, "Robust principal component analysis?," *Journal of the ACM*, vol. 58, no. 3, pp. 11, 2011.
- [13] Kyong Hwan Jin, Dongwook Lee, and Jong Chul Ye, "A general framework for compressed sensing and parallel MRI using annihilating filter based low-rank Hankel matrix," *arXiv preprint arXiv:1504.00532*, 2015.
- [14] Nathan Srebro, *Learning with matrix factorizations*, Ph.D. thesis, Dept. of Elect. Eng. and Comput. Sci., MIT, MA, 2004.
- [15] Stephen Boyd, Neal Parikh, Eric Chu, Borja Peleato, and Jonathan Eckstein, "Distributed optimization and statistical learning via the alternating direction method

of multipliers,” *Found. Trends in Mach. Learn.*, vol. 3, no. 1, pp. 1–122, 2011.

- [16] Marco Signoretto, Volkan Cevher, and Johan AK Suykens, “An SVD-free approach to a class of structured low rank matrix optimization problems with application to system identification,” in *IEEE Conf. on Decision and Control*, 2013, number EPFL-CONF-184990.
- [17] Pascal Getreuer, “Rudin-Osher-Fatemi total variation denoising using split Bregman,” *Image Processing On Line*, vol. 10, 2012.
- [18] Zhouchen Lin, Minming Chen, and Yi Ma, “The augmented lagrange multiplier method for exact recovery of corrupted low-rank matrices,” *arXiv preprint arXiv:1009.5055*, 2010.